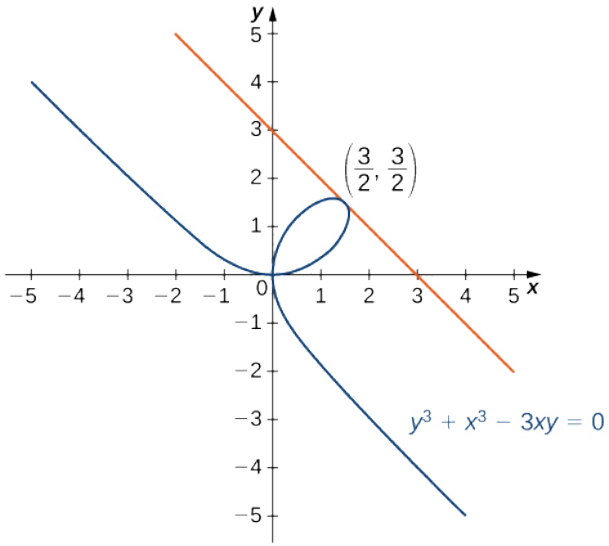
2627-MA140 Engineering Calculus

Week 05, Lecture 2
Implicit Differentiation

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Today, in Engineering Calculus...

1. Implicit Functions
2. Implicit Differentiation
3. Exponential functions
4. Properties of exponents
5. Exercises

Slides are on canvas, and at niallmadden.ie/2627-MA140

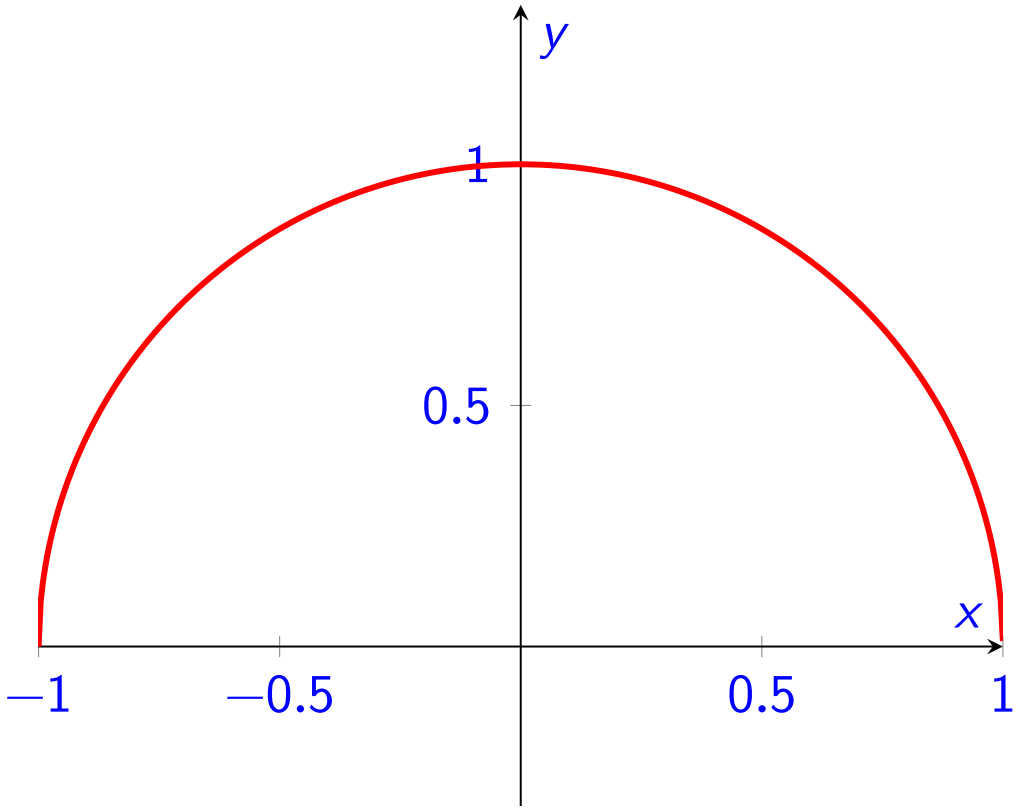


See also: Sections 3.8 (Implicit Differentiation) and 3.9 (Derivatives of Exponential and Logarithmic Functions) of **Calculus** by Strang & Herman: https://openstax.org/books/calculus-volume-1/pages/3-8-implicit-differentiation and https://openstax.org/books/calculus-volume-1/pages/3-9-derivatives-of-exponential-and-logarithmic-functions

Implicit Functions

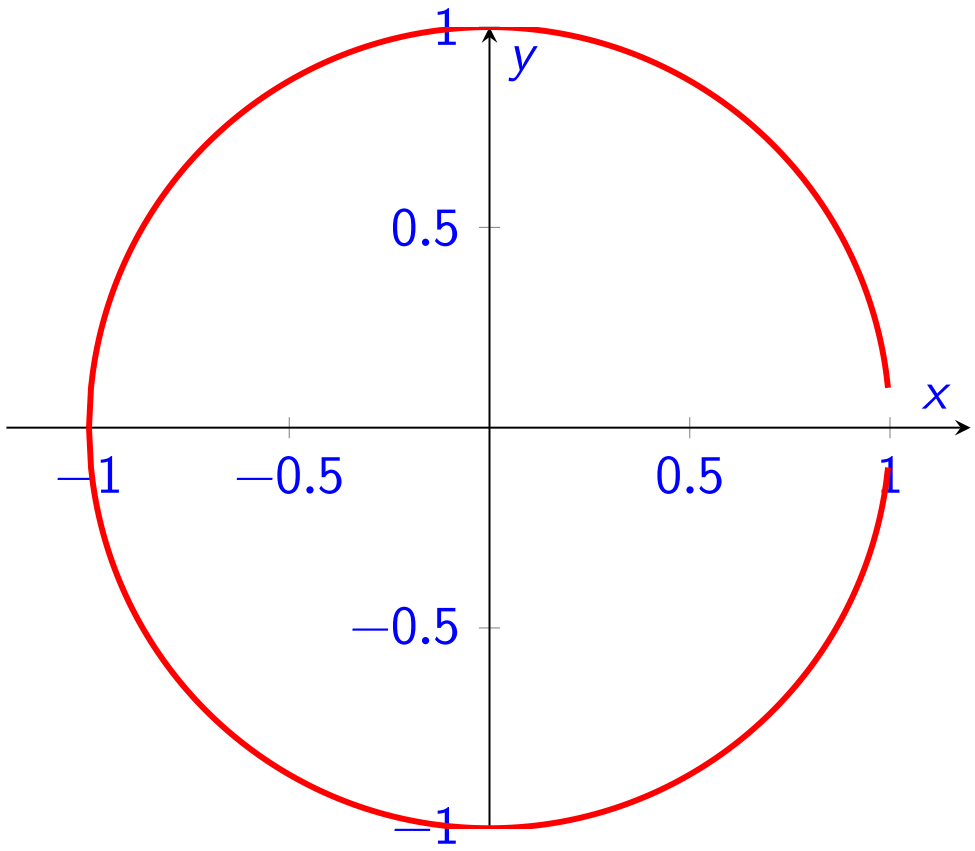
To date, most functions we have studied have been **explicitly** defined. Such functions and be written as $y = f(x)$: given a value of x we can substitute it into $f(x)$ to get the corresponding value of y

Example: $y = \sqrt{1 - x^2}$.



Implicit Functions

However, sometimes we are given an equation involving x and y where these two terms are not "separated" entirely; e.g, $x^2 + y^2 = 1$. Here y is **implicitly** defined: for any pair (x, y) we can check if it is on the curve described by the equation.



Implicit Differentiation

Implicit equations define curves; we can use a technique called **implicit differentiation** to find. tangents to these curves.

Method:

1. Differentiate both size of the equation, with respect to x, keeping in mind that y is a function of x, using the Chain Rule where needed.
2. Solve for dy/dx.

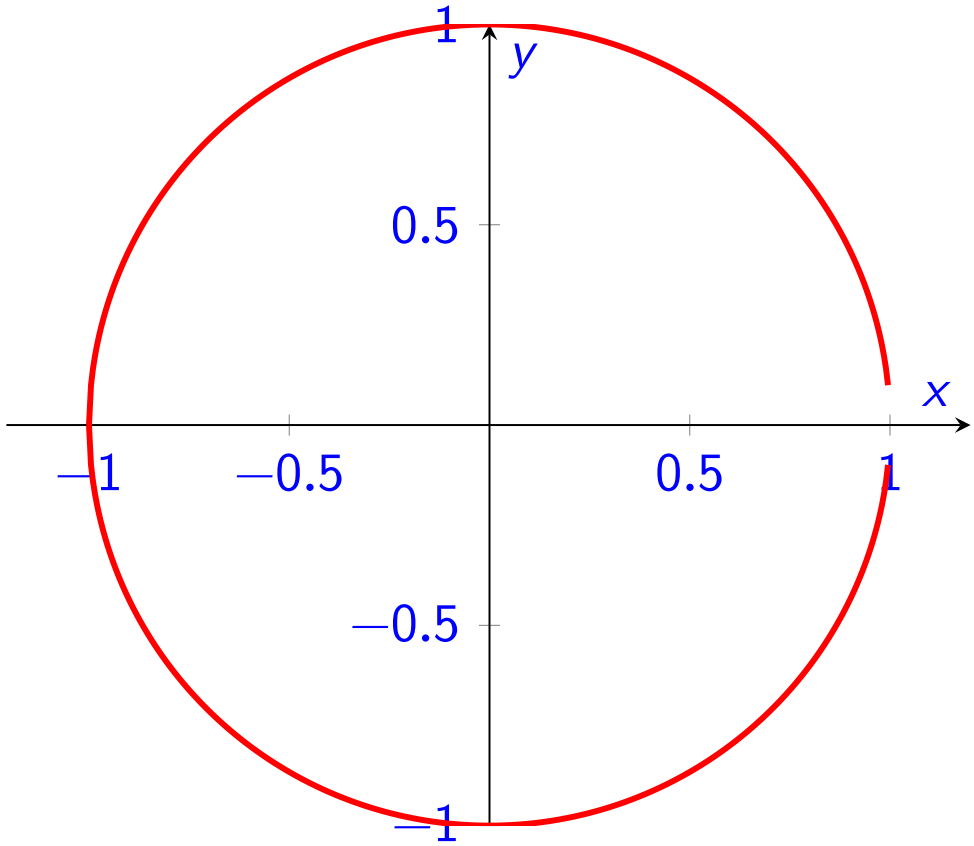
Implicit Differentiation

Example

If y is defined by $x^2 + y^2 = 1$, find $\dfrac{dy}{dx}$.

Implicit Differentiation

Now we know that if $x^2 + y^2 = 1$, then $\frac{dy}{dx} = -\frac{x}{y}$. We can check that this relates to the slope of the tangents to this curve at various places:



Implicit Differentiation

Find the tangent to the curve $x^2+y^2=25$, at the point $(3,-4)$.

First check that the point $(x,y)=(3,-4)$ is on the curve: $(3)^2+(-4)^2=9+16=25$, as required.

Now differentiate:

$$\frac{d}{dx}\left[x^2+y^2\right]=\frac{d}{dx}[25] \quad \Longrightarrow \quad \frac{d}{dx}\left[x^2\right]+\frac{d}{dx}\left[y^2\right]=0$$

$$\Longrightarrow \quad 2x+2y\frac{dy}{dx}=0 \quad \Longrightarrow \quad \frac{dy}{dx}=-\frac{x}{y}.$$

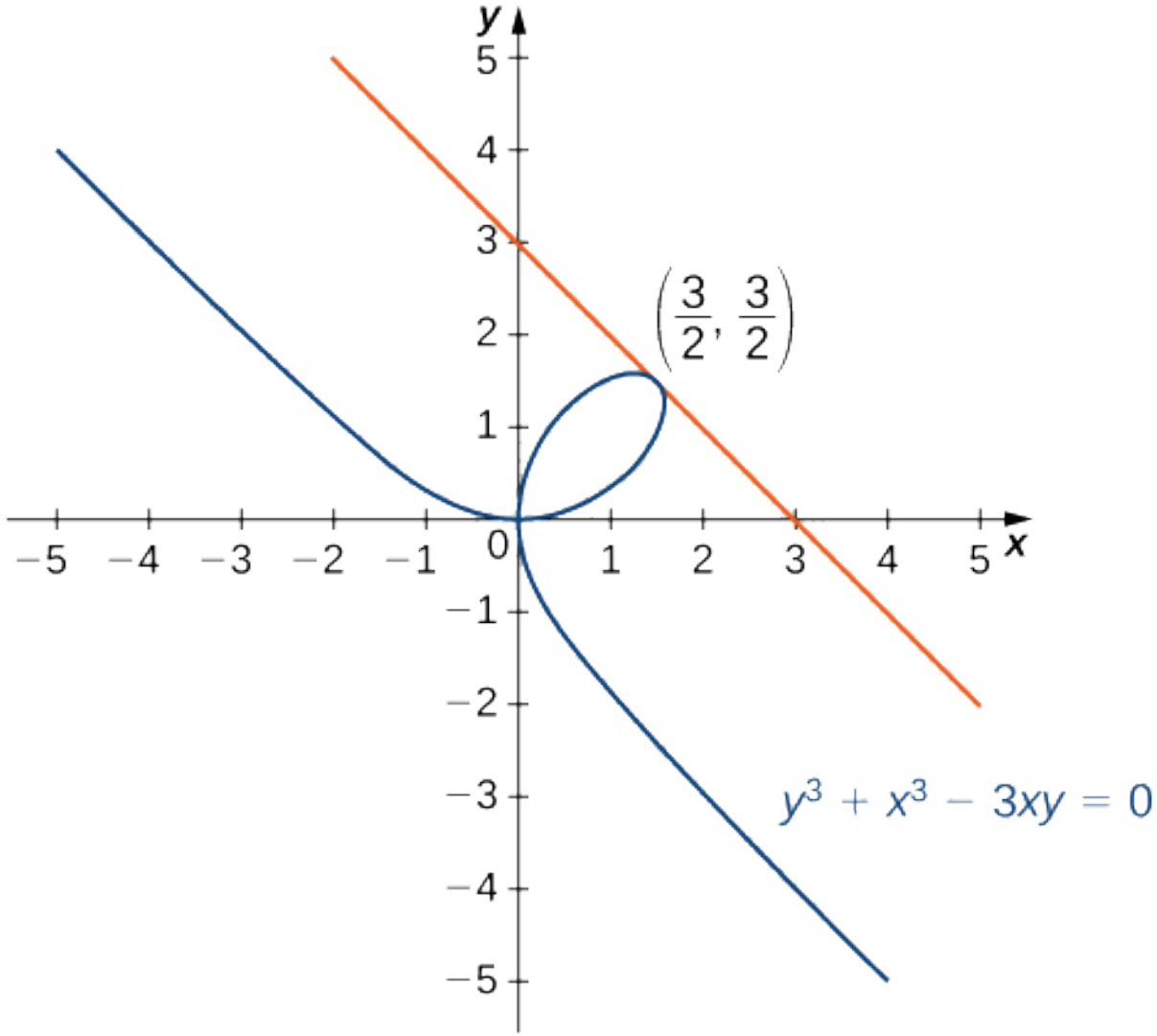
So the slope of the tangent to the curve at $x_1=3$ and $y_1=-4$ is $M=-x/y=3/4$.

The equation of the tangent comes from $y-y_1=M(x-x_i)$. For our values of $x_1=3$, $y_1=-4$ and $M=3/4$, we get $y=(-4)+(3/4)(x-3)$. That is $\boxed{y=\frac{3}{4}x-\frac{25}{4}}$.

Implicit Differentiation

Find the tangent to the curve $y^3 + x^3 - 3xy = 0$, at the point $(3/2, 3/2)$.

Implicit Differentiation



Exponential functions

Earlier in this course we met functions such as $y = x^2$; this is a **power** function.

Now we consider **exponential functions**, such as $y = 2^x$.

Such functions occur in many applications. For example: if I invest €100 with an annual interest rate of 20%, then after x years, I will have €$100 \times (1.2)^x$. **Why?**

Exponential functions

Exponential functions grow quite fast: if my investment is indeed worth $f(x) = 100 \times (1.2)^x$ euros after x years, then...

- After 1 year, I have €120
- After 10 years, I have €619.17
- After 20 years, I have €3,833.80
- After 25 years, I have €9,539.60
- After 50 years, and 190 days, I'll be a millionaire!

Properties of exponents

Here I remind you of some properties of exponents that you should already know: for any positive numbers a and b, and for all x and y,

1. $b^x b^y = b^{x+y}$
2. $\dfrac{b^x}{b^y} = b^{x-y}$
3. $(b^x)^y = b^{xy}$
4. $(ab)^x = a^x b^x$
5. $\left(\dfrac{a}{b}\right)^x = \dfrac{a^x}{b^x}$

There is more information on this in Section 1.5 of the textbook (Exponential and Logarithmic Functions). ($\longleftarrow$ link to book)

Exercises

Exercise 5.2.1

Find the equation of the tangent to the curve defined by $x^2 - y^2 = 16$ at the point $(5, 3)$.

For more exercises on Implicit Differentiation, see https://openstax.org/books/calculus-volume-1/pages/3-8-implicit-differentiation